

COMMENTARY

The distinction between causal, predictive, and descriptive research—there is still room for improvement

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Abstract

It has been proposed that medical research questions can be categorised into three classes: causal, predictive, and descriptive. This distinction was proposed to encourage researchers to think clearly about how study design, analysis, interpretation, and clinical implications should differ according to the type of research question being investigated. This article highlights four common mistakes that remain in observational research regarding the classification of research questions as causal, predictive, or descriptive, and provides suggestions about how they may be rectified. The four common mistakes are (1) Adjustment for “confounders” in predictive and descriptive research, (2) Interpreting “effects” in prediction models, (3) The use of non-specific terminology that does not indicate which class of research question is being investigated, and (4) Prioritising parsimony over confounder adjustment in causal models. © 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Key words: Causal inference; Confounding; Descriptive epidemiology; Prediction; Prognosis; Prognostic factor

1. Introduction

It has been proposed that medical research questions can be categorized into 3 classes—causal, predictive, and descriptive [1–4]. Each type of research question requires different considerations for study design, analysis, interpretation, and clinical implications. Causal research tasks involve understanding the extent to which changes in the exposure cause changes in the outcome on average in a population. Prediction research tasks involve identifying variables that predict some health state or developing a prediction model which may be used to predict some health state. This could involve predicting the future occurrence of disease, the presence of disease (diagnosis), or the clinical course of a disease (prognosis) [1]. Descriptive research tasks involve describing the distribution of disease (or other health-related factors), which may include identifying associations with other variables over which the distribution of disease differs [5,6].

While there is a growing appreciation that research methods need to match the type of question being

answered, there are still common mistakes relating to research question classification that can be improved upon. This article highlights 4 common mistakes that remain in observational research regarding the classification of research questions as causal, predictive, or descriptive.

2. Mistake 1: adjustment for “confounders” in predictive and descriptive research

Confounders are accounted for in causal research to reduce bias in the causal effect estimate. Directed acyclic graphs (DAGs) are commonly used to identify confounders and to avoid adjustment for some variables that may induce bias, such as colliders and mediators [7]. The backdoor criterion may then be used to evaluate whether the measured set of confounders is sufficient to account for (assuming the DAG is correctly specified) [8]. If the set of confounders satisfies the backdoor criterion and is adequately accounted for through study design and analysis, then this would sufficiently eliminate confounding (and avoid some types of overadjustment bias) [7,9].

Confounding is a causal concept, but is frequently inappropriately referred to in prognostic factor research [10]. Prognostic factor research aims to identify variables that are associated with a future health outcome among people

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What is new?

Key findings

- This commentary describes 4 common mistakes relating to the classification of data science tasks as causal, predictive, or descriptive. The 4 mistakes described are (1) adjustment for “confounders” in predictive and descriptive research, (2) interpreting “effects” in prediction models, (3) the use of nonspecific terminology that does not indicate which class of research question is being investigated, and (4) prioritizing parsimony over confounder adjustment in causal models.

What this adds to what is known

- The proposal that data science tasks can be classified as either causal, predictive, or descriptive has helped researchers appreciate that study design, analysis, interpretation, and clinical implications should differ according to which of the 3 tasks is being conducted. This article highlights four areas for improvement, relating to the distinction between data science tasks in medical research.

What is the implication and what should change now?

- Researchers should avoid referring to variables as “confounders” in descriptive or predictive research, since confounding is a causal concept.
- Causal risk reduction should not be inferred from prediction models, and the word “effect” should not be used to describe the associations of predictors with the outcome in prediction research.
- Journals, reviewers, and reporting guidelines should require authors to specify whether their research question relates to causality, prediction, or description.
- When it is unclear whether a variable is a confounder, sensitivity analysis should be conducted to assess the impact of including/excluding the variable from the adjustment set.

with a particular disease or health status [11,12] (For those not familiar, a thorough introduction to prognostic factor research is provided in the PROGRESS framework [11]). While an unadjusted association is appropriate to assess evidence of whether a variable is a prognostic factor, it is also recommend that studies present an adjusted estimate where established prognostic factors have been adjusted for [11,13]. The purpose of the adjusted analysis is to assess evidence of whether, and the extent to which, the candidate

prognostic factor may be a predictor over and above the established prognostic factors [12,13], that is, to assess evidence of whether the association is still present after adjusting for the established prognostic factors and assessing the magnitude of that adjusted association. If a prognostic factor is a predictor of the outcome univariably, but is strongly correlated with an established prognostic factor, then it may be the case that the new prognostic factor is not a predictor over and above the established prognostic factors. For example, bulbar onset and age are associated with poorer prognosis in amyotrophic lateral sclerosis, but amyotrophic lateral sclerosis onset also occurs at a later age for bulbar onset [14]. It was therefore important for researchers to confirm that bulbar onset is predictive of poorer prognosis over and above age in adjusted analyses [14] (Note that this approach to adjustment in prognostic factor studies differs from the discouraged practice of using data-driven techniques to identify so-called “independent predictors” from a list of candidate predictors [13]. In this recommended approach, only well-established predictors are adjusted for, and no variable reduction techniques are used).

If there is an intention to intervene on the prognostic factor to improve health outcomes, then the research question is causal and not predictive. Such research would need to adjust for confounders to estimate the causal effect of the prognostic factor on the outcome (or use other appropriate methods, such as randomization) to unbiasedly estimate a causal effect [15].

Descriptive epidemiology also suffers from the incorrect belief that “confounders” should be accounted for [5]. In descriptive research, it may be helpful to characterize the distribution of the outcome being described according to additional variables that serve to stratify the larger target population; however, these variables should not be referred to as confounders, as confounding is a specific concept relevant to causal questions. For example, a researcher may wish to describe COVID-19 mortality rates (a measure of the outcome) in the UK (the target population) according to age groups, sex, or ethnicity (strata of the larger population). Such research would help to identify subgroups of the UK population that could be at a higher risk of dying from COVID-19 to target resources (eg, vaccines) or to generate hypotheses for causal or predictive research. However, in descriptive research, simultaneously adjusting for a large number of variables in a single model can lead to results that are difficult to interpret. For example, a researcher may use a linear mixed-effects model to describe average patient trajectories for an outcome over time. The variable sex may then be introduced into the model to estimate the average difference in the outcome between men and women during the study’s follow-up period. However, if numerous additional covariates are added to the model, then the sex coefficient would indicate the average difference between men and women during the study’s follow-

up period, while holding all other covariates constant, which would be unlikely to correspond to any well-defined descriptive question. Such an approach usually indicates that the research methods were mistakenly halfway between causal and descriptive, with the researcher believing that they had to “adjust for confounders”, as is done in causal research. Instead, accounting for additional variables in descriptive research should be guided by the research question, with careful thought about which population and strata of that population are being described [5].

3. Mistake 2: interpreting “effects” in prediction models

The term “effect” is commonly used in prediction research, but predictors do not necessarily need to have a causal effect on the outcome to be good predictors. Variables that do cause the outcome will always be predictors (even if they are not strongly predictive), but it is not necessary for a predictor to be a cause. For example, tumor markers are predictors of cancer progression and recurrence; however, a hypothetical intervention to reduce tumor markers would not cause an improvement in cancer outcomes [16]. A 2021 scoping review found that in 30% of prediction modeling studies, coefficients were mistakenly interpreted causally [10]. That is, an individual may incorrectly think that if they calculate the probability of the outcome using a prediction model before and after changing the value of one of the predictors, then this would indicate the amount an individual could reduce their risk of the outcome if they were to change their lifestyle in this way. Prediction models are not designed to estimate causal effects, so interpreting the model in this way can produce highly misleading results. To avoid this mistake, the term “effect” should be avoided when describing the associations in prediction models. Some statisticians may argue that the word “effect” does not necessarily imply causality among statisticians, but it is best to avoid the term when referring to noncausal associations, given that many epidemiologists believe the word implies causality [17].

Prediction models aim to use multiple predictors to collectively estimate the absolute risk or value of the outcome, but the individual coefficients are not particularly meaningful beyond being used to estimate the predicted risk/value of the outcome. Thus, care should be taken to avoid straying away from the intended use of the model. Authors should avoid presenting model coefficients in a way that may invite causal interpretation, such as reporting transformed coefficients, such as odds ratios or hazard ratios, or reporting the corresponding *P* values. Untransformed coefficients, such as log odds ratios or log hazard ratios, may be less likely to be incorrectly interpreted in such a way.

4. Mistake 3: nonspecific terminology

The growing appreciation that different research questions require different methods and interpretations has highlighted issues with some traditional epidemiologic terminology. Authors have argued that the term risk factor is a misleading term since it has frequently been used to mean either a cause of the outcome or a predictor of an outcome [18]. In their 2016 *BMJ* article, Huitfeldt highlights the issue of using nonspecific terminology, where they highlight that the answer to the question “Is caviar a risk factor for being a millionaire?” is entirely dependent on whether the definition of the term “risk factor” is in the causal sense or the prediction sense [18]. Researchers can use terms such as etiological/causal risk factor or predictor, predictive risk factor, prognostic factor, or risk marker/indicator to distinguish between studies aiming to estimate a causal effect from those identifying predictors. Similarly, other terms like “association” often need clarification regarding the type of association being investigated. Ambiguous language makes it difficult to critically appraise a study and to understand the clinical implications of the research.

Historically, it has been suggested that causal language should not be used when discussing the results of an observational study. Hernán argued that causal language is necessary to adequately describe the study’s aim and that authors should explicitly state that they seek to answer a causal question [19]. The Journal of the American Medical Association framework for causal inference in medical and health policy research also supports the argument that banning causal language makes it difficult for authors to adequately describe their research goal [20]. The framework also disputes the argument for a blanket ban on causal language when interpreting the results of observational research and instead suggests a “context-informed examination of all relevant aspects of design, conduct, and analysis” [20].

There is a need to use language that makes it easier for readers to understand the context of the research, including specifying whether each research question is causal, predictive, or descriptive. To facilitate this, reporting guidelines, such as the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement [21], should require researchers to state which type of research question is being investigated for each study aim. Reviewers and journals should also encourage authors to clarify this in their articles.

5. Mistake 4: prioritizing parsimony over confounder adjustment in causal models

Sometimes the causal role of a variable in a DAG will be unclear, making it difficult to determine whether the variable is a confounder. Researchers may be tempted not to include such variables, under the belief that they are building a “parsimonious model”. Parsimony may be a helpful

concept to consider in prediction modeling to reduce overfitting (as long as it is not at the expense of predictive accuracy). The term is not commonly used in the causal inference literature, though, since the main focus of variable selection is often on reducing bias to estimate a causal effect.

In causal research, in such situations where the causal role of a variable in a DAG is unclear and a researcher is unsure whether the variable is a confounder, careful thought should be given to weigh up the potential implications of adjusting for the variable if it isn't a confounder, vs potentially missing a confounder. DAGs can be helpful for this purpose, in particular when they are arranged in temporal order, as recommended by Tennant et al [22]. Post-treatment variables have the potential to induce bias if conditioned on, either through blocking causal pathways or also by inducing collider bias; thus, adjustment should be avoided, and if there is debate about whether the variable is a confounder or mediator then a sensitivity analysis with and without the variable is encouraged, along with cautious interpretation. However, for pretreatment variables, there is often less to be worried about.

Adjustment for pretreatment variables that are not confounders will often not induce bias. The first scenario where bias could be introduced is the scenario of "M-bias", where collider bias is induced by opening a back-door path between the exposure and outcome; however, the M structure is rare in practice [23]. The second scenario is that adjusting for an instrumental variable may introduce "Z-bias" [24] (See Liu et al [23] and Myers et al [24] for detailed explanations of M-bias and Z-bias.). In both scenarios, it is recommended that, if in doubt, controlling for confounding should take precedence since it can lead to more bias than M-bias and Z-bias [23,24]. Thus, if there is ambiguity surrounding whether a pretreatment variable is a confounder, purely a cause of the exposure, or purely a cause of the outcome, then there is often little risk of inducing bias unless 1 of the 2 aforementioned structures is present. In the 2 scenarios where bias can be introduced, it is recommended that avoiding unmeasured confounding should take precedence. Even though the risk of bias when adjusting for pretreatment variables is often low, sensitivity analysis with and without adjustment for debatable confounders should be conducted. Authors should not be tempted to think that it is wise to avoid a potential confounder if they are unsure about whether it truly is a confounder. Careful consideration should be given to the implications of adjustment, and sensitivity analysis should be conducted, along with discussing the potential implications on the causal effect estimate of including/excluding the variable from the adjustment set. Sensitivity analyses should be presented alongside prespecified interpretations of why the results may differ, which should be arrived at following careful discussions between clinical experts and researchers with expertise in causal inference.

Also, note that adjustment for pretreatment variables can affect precision. Whether precision increases or decreases depends on the covariate role and model type [25]; however, in most observational studies, assuming a sufficiently large sample size, precision is often a lower priority than residual confounding. It is also worth noting that causal effects may be biased when there are inadequate participant numbers for some combinations of the exposure, outcome, and covariates [26]. Greenland et al suggest strategies to detect and address what they call "sparse data bias", but warn that dropping important confounders can worsen bias [26].

The cautious approach of including pretreatment variables that might (but are not known to) be confounders is encouraged in the literature. VanderWeele argues that, when there is incomplete knowledge of a causal diagram, researchers should adjust for all covariates that cause the exposure (with the exception of instrumental variables), cause the outcome, or cause both [27]. Tennant et al argue that arcs/arrows in DAGs should generally be assumed to exist and only be excluded if there is evidence or logic for the absence of a causal effect [22].

6. Conclusion

This article summarizes 4 mistakes that are still common, which relate to conflation between causal inference, prediction, and descriptive research (but is not a comprehensive list). Tailoring methods toward the type of research question and carefully interpreting results in the appropriate context can improve the quality of medical research, so it should be encouraged by all those involved in medical research and biostatistics/epidemiology teaching.

Declaration of competing interest

There are no competing interests for the author.

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Data availability

No data was used for the research described in the article.

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